Date: 23-07-2026

Cisco Packet Tracer

AIM:
To study the packet tracer tool installation and user interface

Description:-
This experiment introduces the Cisco Packet Tracer Simulation tool and it's user interface for designing and simulation computer networks. It also covers the basics networking devices such as hubs and switches A hub broadcasts data to all connected devices, whereas a switch forwards data only to the intended destination using MAC addresses, making communication faster and more efficient.

Procedure:-
1. Install & launch Cisco Packet Tracer.
2. observe the user interface, including the device toolbar and workspace.
3. Drag and place networking devices such as PCs, hubs, and switches onto the workspace.
4. connect the devices using the appropriate cables.
5. Assign IP addresses to the PCs.
6. Test connectivity between the devices using

the ping command.

7. Observe the data transmission through the hub and the switch.

8. Save the network topology and record the observations.

Observation:-

a) Behavior of switch and hub in forwarding Packets.

Hub: A hub forwards the received data to all connected devices without checking the destination address.

Switch: A switch forwards the received data only to the destination device by using the MAC address, making communication faster and more secure.

b) Network topology implemented in the college.

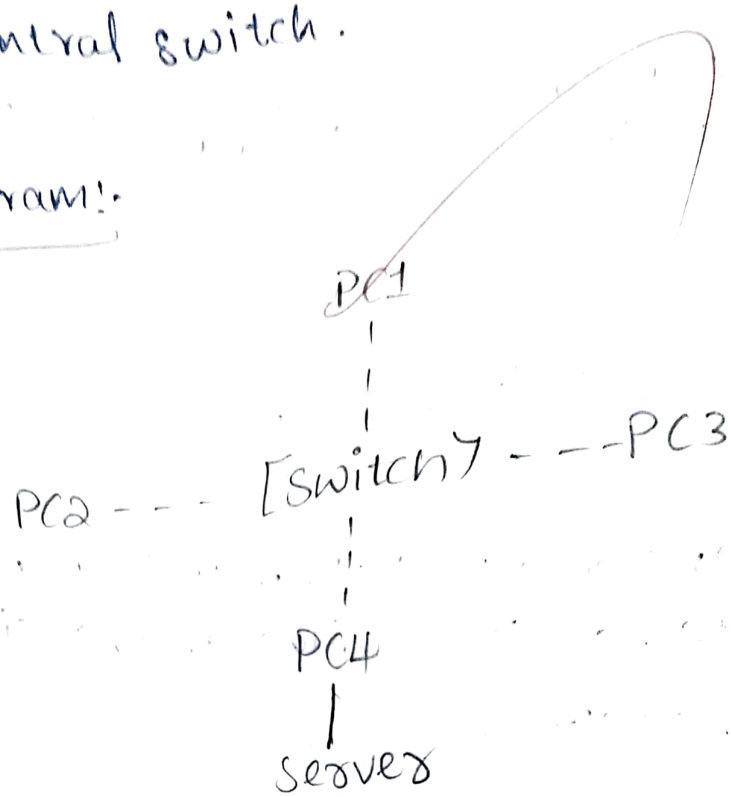
Most colleges, use a star Topology, where all computers are connected to a central switch.

* The hub broadcasts packets to all connected devices.

* The switch forwards packets only to the intended destination using MAC addresses.

* The network topology observed is star Topology, with all devices connected through a central switch.

Diagram:-



Result:-

The cisco packet Tracer environment was successfully explored. The working of hubs